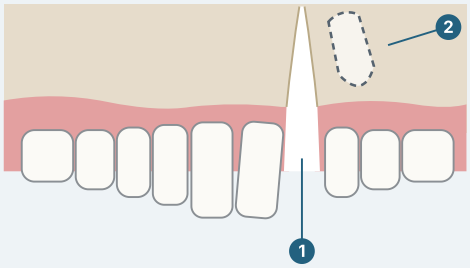
Volunteers Needed · Research Study

2D or 3D? Understanding bone grafts for a cleft in the gum

Study title: Seeing the stages: comparing 2D, augmented reality, video and 3D printed education materials to explain alveolar bone grafting in cleft lip and palate.



1 Gap in the gum bone (alveolar cleft)
2 Adult eye tooth (canine) waiting in the bone

Simplified drawing for illustration only.

What is an alveolar bone graft?

Some children are born with a cleft lip and palate. Surgery in early life closes the lip and the roof of the mouth, but a gap often stays in the bone that holds the teeth. This gap is the alveolar cleft.

At about 8 to 11 years old, a surgeon fills the gap with bone, most often from the hip. This gives the adult eye tooth bone to come down into. Treatment goes on for years, so this study shows it in three stages: before surgery, about 3 months after, and years later.

About this study

This project is part of a PhD at Nagoya University, with Professor Toko Hayakawa and Professor Nagato Natsume of Aichi Gakuin University. The researcher is a dentist who is interested in how people rate 2D and 3D materials for understanding this treatment. Each participant looks at four ways of explaining the three stages and then rates and ranks them.

2D information sheet · **AR web page on an iPad (augmented reality)** · **3D printed model** · **Narrated video**

Icons only. The actual study materials are shown for the first time during the study.

Who can take part?

People aged 18 or over, in any of these groups:

- Dental students
- Medical students
- Dentists
- Medical doctors
- Oral and maxillofacial surgeons
- Other health care workers (for example nurses, speech pathologists, dental hygienists)
- Members of the public (no health care training)

No experience of cleft lip and palate is needed to take part.

What does taking part involve?

1. Scan the QR code, read the information and give consent online.
2. Look at the four explanations one at a time, for a short, fixed time each. Each one shows all three stages.
3. Answer short questions about each one, then rank all four.
4. The whole session takes about [X] minutes.

Good to know

- Taking part is voluntary. Stopping is possible at any time before submitting.
- No names are collected. Answers are anonymous and kept on Nagoya University secure storage.
- There is no cost to take part.
- This study does not give personal medical or dental advice.

PLACEHOLDER QR CODE

Scan the QR code to read the information and consent form (Microsoft Forms).

For more information, please contact:
seifert.markus.luke.f5@s.mail.nagoya-u.ac.jp